Leo's Big Box of Numbers



100





Counting to 100 together.

Teacher Note: This visual guide uses Singapore Math principles (Concrete-Pictorial-Abstract, Number Bonds, and Place Value) to build a lasting number foundation.

Leo has a lot of toys.

Counting them one by one takes a long time!

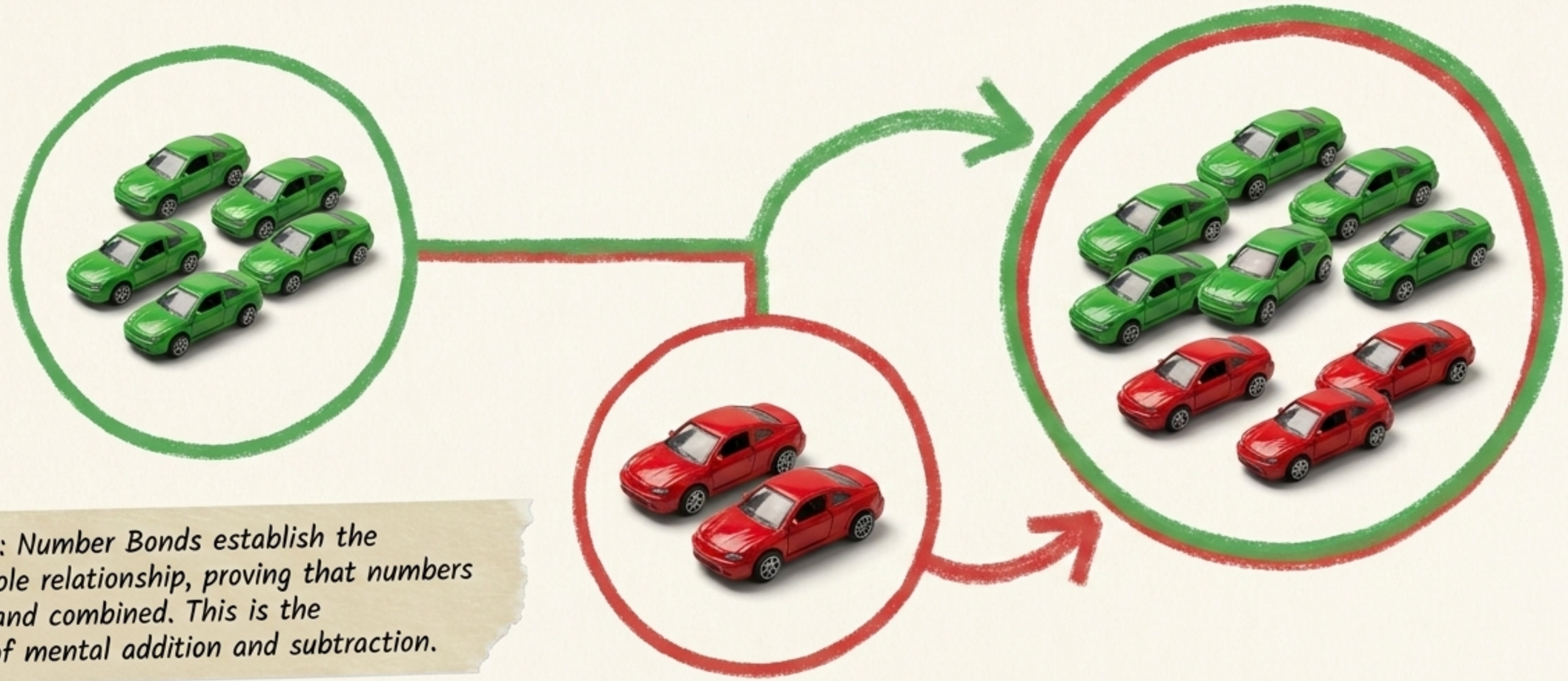




Teacher Note: Young learners often struggle with one-to-one correspondence for large quantities. This sets the stage for teaching grouping as a more efficient counting strategy.

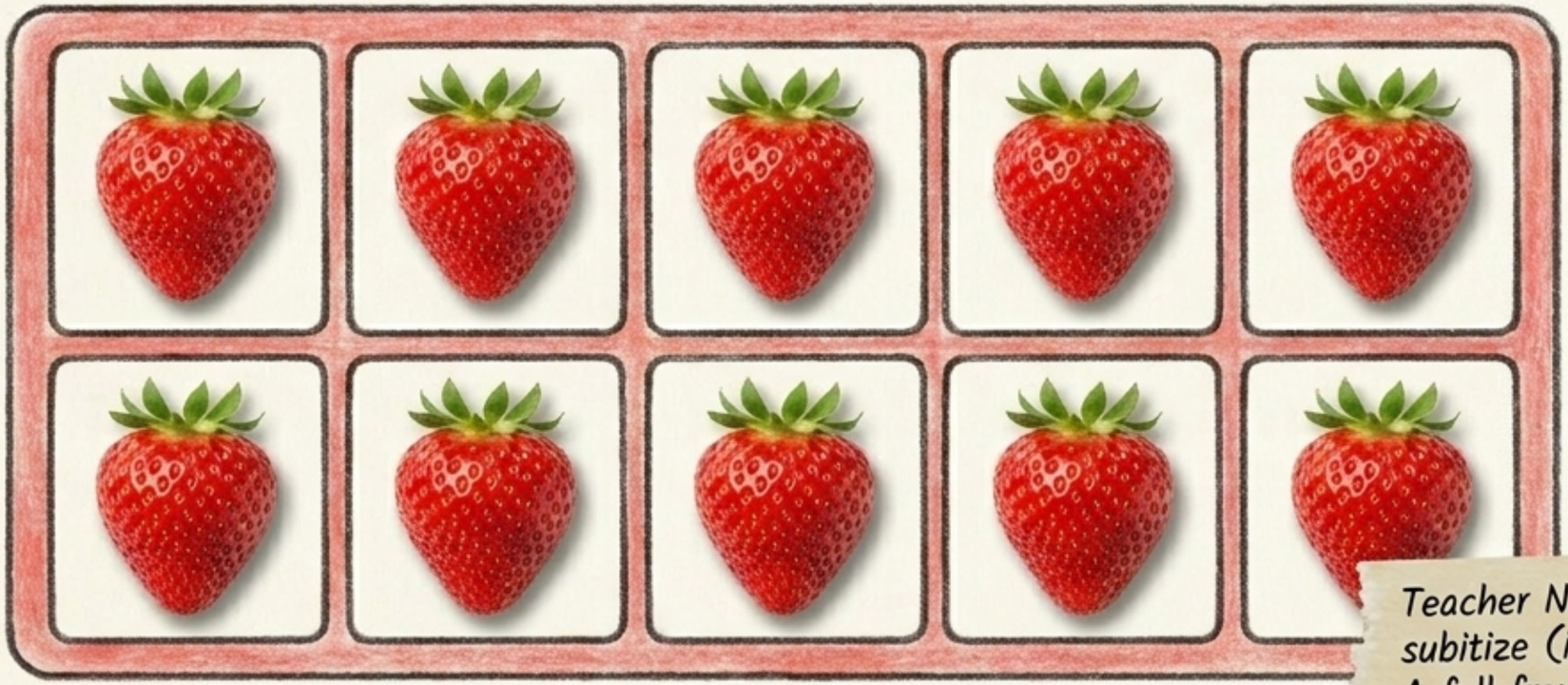
Numbers are made of parts.

7 cars and 3 cars make 10 cars!



Teacher Note: Number Bonds establish the part-part-whole relationship, proving that numbers can be split and combined. This is the cornerstone of mental addition and subtraction.

When the frame is full, we have one perfect group of TEN.



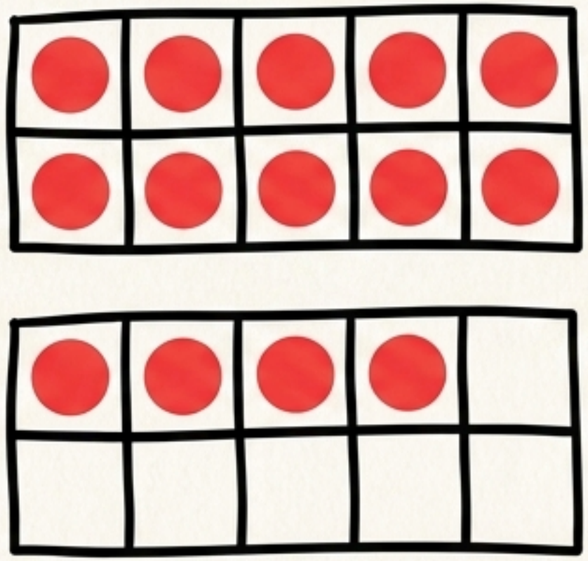
Teacher Note: The Ten-Frame helps children subitize (instantly recognize) numbers to 10. A full frame anchors "10" as a single structural unit, preparing them for place value.

What happens after 10? We add some more!

Left Panel (Concrete)



Pictorial (Pictorial)

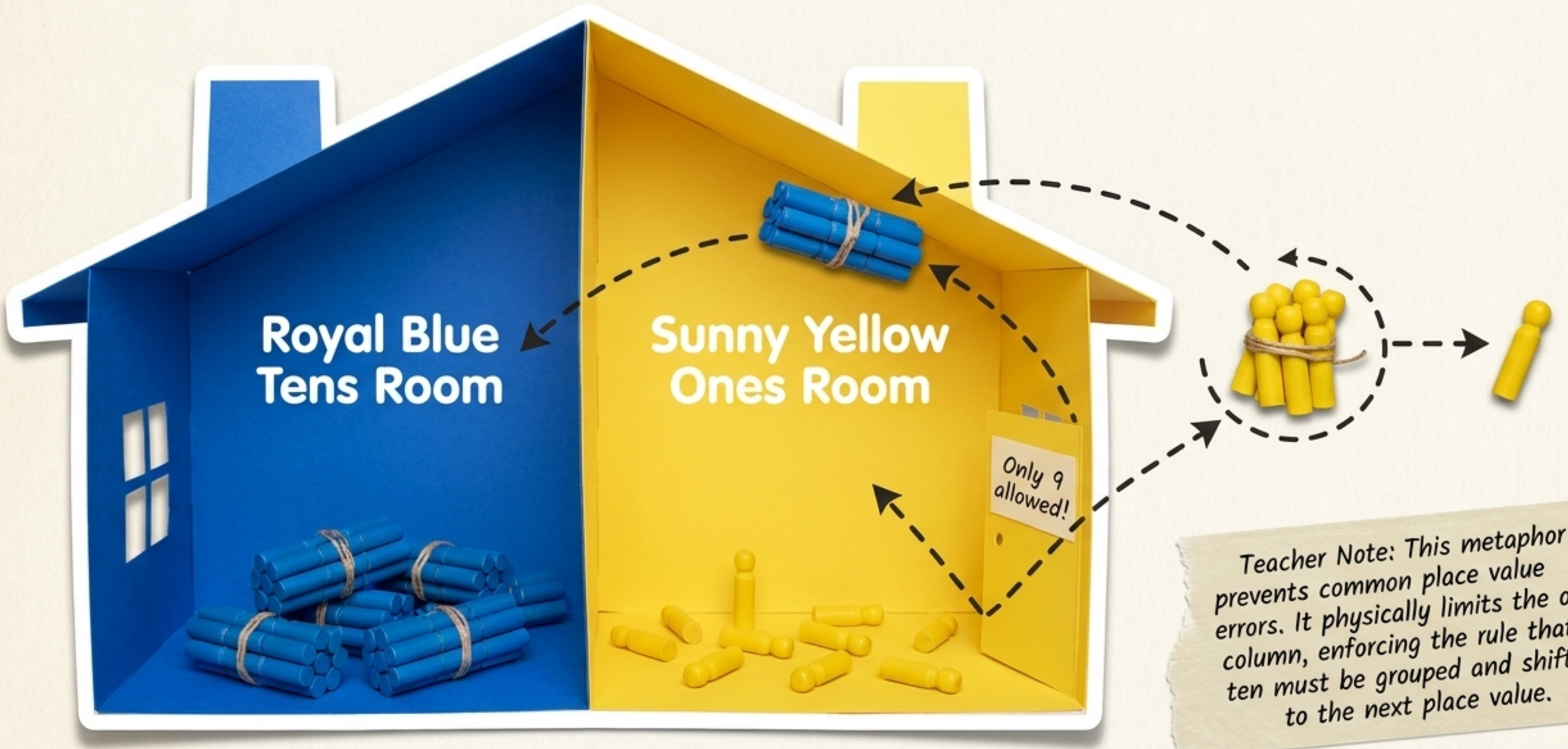


Right Panel (Abstract)

$$10 + 4 = 14$$

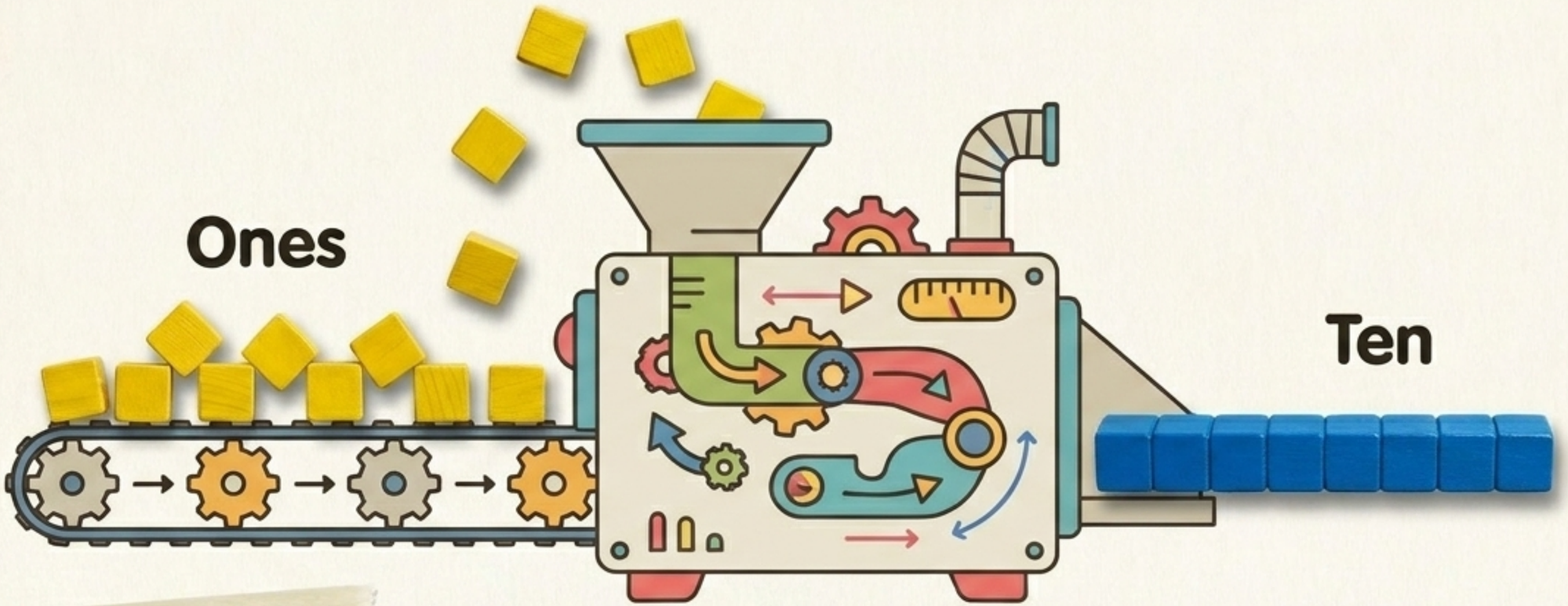
Teacher Note: The Concrete-Pictorial-Abstract (CPA) approach bridges the gap between physical reality and mathematical symbols. Teen numbers are simply ten and some ones.

The Ones Room can only hold 9. When we get 10, they pack up and move to the Tens Room!



Teacher Note: This metaphor prevents common place value errors. It physically limits the ones column, enforcing the rule that a ten must be grouped and shifted to the next place value.

10 loose blocks become 1 solid stick.



Teacher Note: Understanding that 'ten' is simultaneously ten individual things AND one single group is the most vital cognitive leap in first-grade mathematics.

10 Ones = 1 Ten

Instead of counting one crayon at a time, Leo counts whole boxes!

Ten... Twenty... Thirty!



Teacher Note: Skip counting by 10s is the foundation of multiplication and efficient counting. We color-code the tens in Royal Blue to maintain visual consistency.

Counting by tens is super fast.



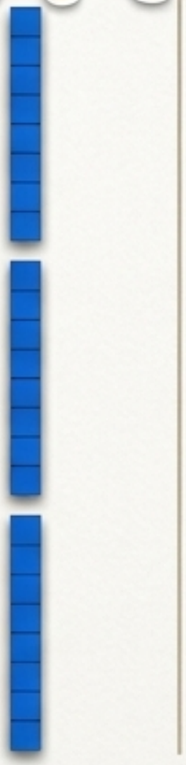
Race Track
The Turtle Track



Race Track
The Cheetah Track

1, 2, 3, 4, 5...
this takes
forever!

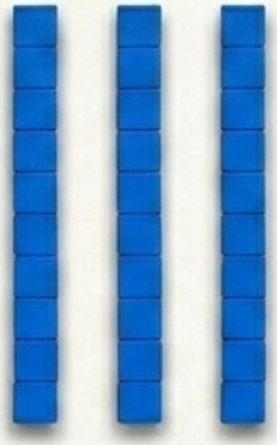




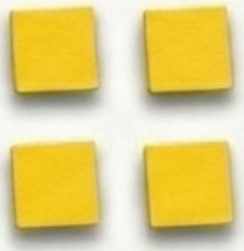
10, 20, 30, 40.
Done!

Teacher Note: This matrix demonstrates efficiency. Students often resist grouping until they visually recognize how much time and effort it saves them.

Thirty-Four is just 3 bundles of ten, and 4 extra ones.



3 Tens

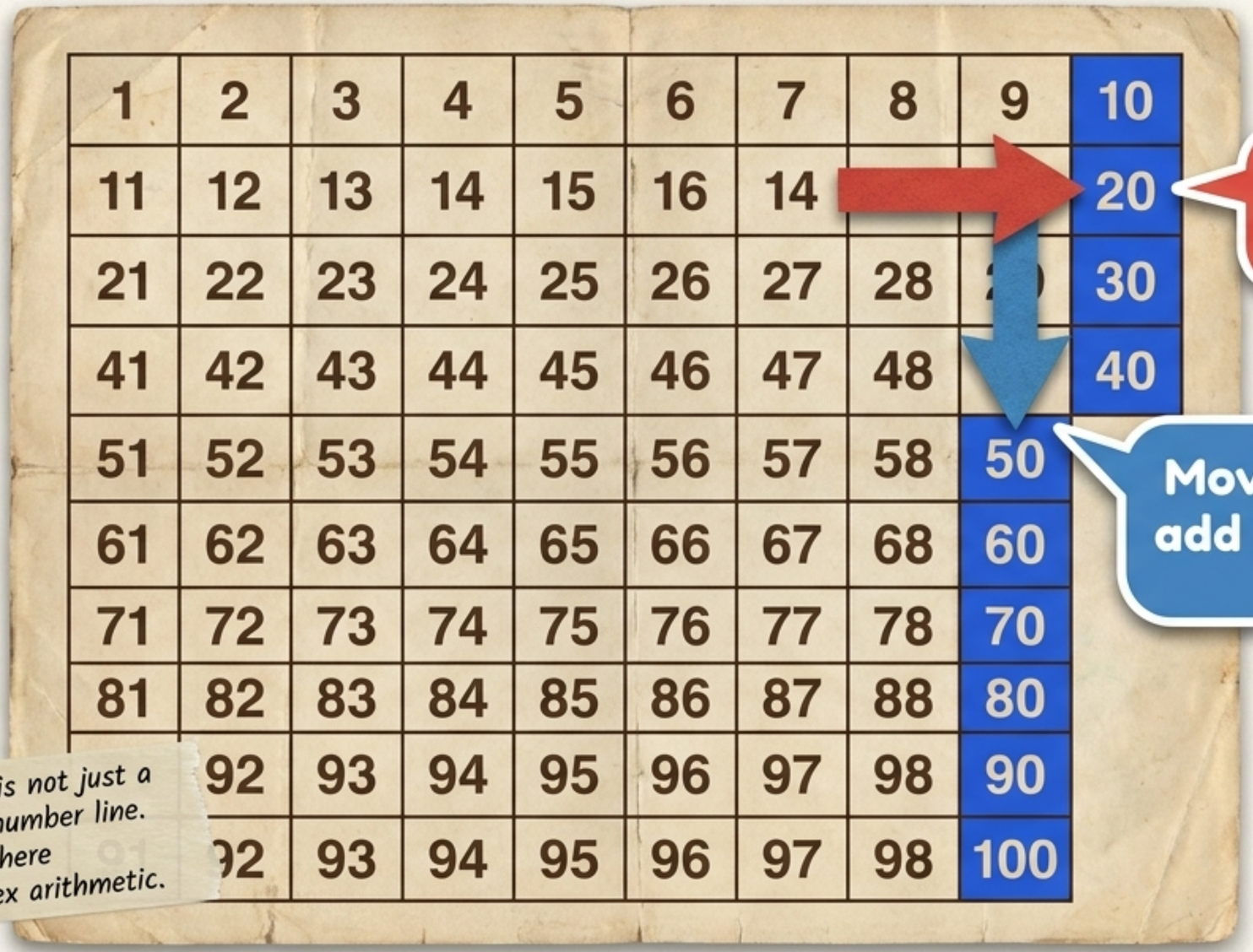


4 Ones

30 + 4 = 34

Teacher Note: By consistently treating 2-digit numbers as decomposable parts (tens and ones), children build a robust number sense that naturally doeirts euch that naturally supports double-digit addition later.

The 100-Chart is a map full of secrets!



Teacher Note: The 100-chart is not just a list; it is a two-dimensional number line. Recognizing spatial patterns here prepares students for complex arithmetic.

One Hundred is just 10 groups of 10. Leo did it!



Teacher Note: 100 is an abstractly large number for a 6-year-old. Visually proving that it is simply ten tens makes the concept accessible, manageable, and wonderfully clear.